<table>
<tr><td>7a(i)</td><td>Faraday’s Law of Electromagnetic Induction states that the <u>induced e.m.f</u> is directly <u>proportional to the rate of change of magnetic flux linkage</u>.</td><td>B1</td></tr>
<tr><td>7a(ii)</td><td>A (primary) coil is installed in the charging plate and a another (secondary) coil in the device.

A <u>changing</u> magnetic flux is produced in the (primary) coil of the charging plate, due to the alternating current from the power source.

By Faraday’s law, an emf will be induced in the (secondary) coil of the device which will charge the device.</td><td>B1

B1

B1</td></tr>
<tr><td>7a(iii)</td><td>Advantage: (any valid suggestion)
• No connecting wires required between device and charging plate, so less risk of electrical shock.
• Can fully enclose the charging parts to make it water-proof.

Disadvantage: (any valid suggestion)
• Coils of wires are required to be installed inside the device, hence more bulky/costly.
• Less efficient than wired connection due to resistive heating (Lenz’s law).
• Longer charging time.
• Device will need to stay with the charging plate and is not mobile when charging.</td><td>B1

B1</td></tr>
<tr><td>7b(i)</td><td>$2\pi f = 377$
$f = \frac{377}{2\pi}$
$= 60.0\text{ Hz}$</td><td>C1
A1</td></tr>
<tr><td>7b(ii)</td><td>$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{340}{\sqrt{2}}$
$= 240\text{ V}$</td><td>C1
A1</td></tr>
<tr><td>7b(iii)</td><td>It means that this alternating supply voltage will <u>provide the same average/mean power (or rate of heat dissipation)</u> as an equivalent <u>steady direct voltage of 240 V</u>.</td><td>B1</td></tr>
<tr><td>7c(i)</td><td>$\varepsilon = BLv$
$= (0.54)(0.100)(2.3)$
$= 0.1242 = 0.124\text{ V}$</td><td>C1
A1</td></tr>
<tr><td>7c(ii)</td><td>end C</td><td>B1</td></tr>
<tr><td>7c(iii)</td><td>$I = \frac{\varepsilon}{R} = \frac{0.1242}{0.83}$
$= 0.1496 = 0.150\text{ A}$</td><td>C1
A1</td></tr>
<tr><td>7c(iv)</td><td>By Fleming’s LHR, since current flows from A to B and magnetic field is into the plane,
so force on rod AB is towards right.</td><td>M1
A1</td></tr>
<tr><td>7c(v)</td><td>$F = BIL = (0.54)(0.1496)(0.100)$
$= 8.08 \times 10^{-3}\text{ N}$</td><td>C1</td></tr>
</table>

	$a = \frac{F}{m} = \frac{8.08 \times 10^{-3}}{0.034}$ $= 0.238 \text{ m s}^{-2}$ [A1]	A1
	Total:	20